

Data Transformation Pipeline

ETL Visual Architecture Blueprint — Power Query to Looker Studio

1. Structural Flow Overview

This diagram visualizes how raw, separate data tables are combined and cleaned using Power Query, then structured into optimized reporting files ready for the interactive dashboards.

[Stage 1] Relational Data Extraction (Raw Ingestion)

Ingesting 5 tabular flat-files from source directory: Patients.csv, Doctors.csv, Appointments.csv, Treatments.csv, Billing.csv.



[Stage 2] Power Query Core Transformations & Normalization

Enforcing structural alignment: promotions of row headers, dynamic datetime and absolute numeric data-type setting, conditional runtime generation of age cohorts, and targeted denormalizing multi-table Left Outer joins.



[Stage 3] Parallel Output Split Matrix

operational_analytics.xlsx

Primary Target: Operational KPIs

Key Contents:

- Appointments base dataset
- Patient Demographics
- Age Group categorization
- Staff branch mapping

financial_analytics.xlsx

Primary Target: Financial KPIs

Key Contents:

- Billing baseline matrix
- Treatment cost mapping
- Appointment validity links
- Revenue lifecycle logs

clinical_analytics.xlsx

Primary Target: Clinical KPIs

Key Contents:

- Treatments timeline logs
- Physician specialization mapping
- Patient gender splits
- Clinical Age Groups



[Stage 4] Looker Studio Presentation Layer

Connecting processed spreadsheets as distinct dynamic data sources to populate specialized analytical reporting view tabs.

2. Pipeline Execution Logic

By handling all data cleaning, calculations, and connections early in the source files, we minimize processing lag. This ensures the dashboards load instantly when users click or change filters.